41107 Marine and Ocean Engineering

Ships: Motion Dynamics, Hydrostatics, Strength and Resistance

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects the reviewed questions belonging to the **Ships** block. It includes **ship motion dynamics**, **encounter frequency / forward speed**, **Froude-Krylov assumptions**, **ship hydrostatics**, **hydrostatic coefficients**, **strength of ships** and **ship resistance / model testing**. Exact duplicates across the reviewed MCQ documents were removed. Same pattern with different numerical values or different answer options is kept as a separate variant.

The questions are ordered as follows:

1. Ship motion dynamics, encounter frequency and Response Amplitude Operator (RAO).
2. Ship hydrostatics, stability, trim and hydrostatic coefficients.
3. Strength of ships.
4. Ship resistance, model testing and Froude scaling.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants and notation

$$g = 9.81\ \mathrm{m/s^2}, \qquad \rho_{\mathrm{water}} \approx 1025\ \mathrm{kg/m^3}, \qquad 1\ \text{knot} = 0.51444\ \mathrm{m/s}$$

$$\omega = \frac{2\pi}{T} = 2\pi f, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi}, \qquad \text{freshwater: } 1\ \mathrm{m^3} \approx 1\ \text{tonne}$$

Contents

Ship motion dynamics, encounter frequency and Response Amplitude Operator (RAO)

M1. Pitch transfer-function phase

SHIP MOTION DYNAMICS: Transfer functions of a conventional container ship are available. At a given frequency $\omega = 0.6$ rad/s, the real and imaginary parts of the pitch transfer function is $(\mathrm{Re}, \mathrm{Im}) = (-5.0 \times 10^{-2},\ 8.0 \times 10^{-2})$ rad/m for wave heading 150°. What is the phase lag ϕ relative to the wave?

Options:

A. $\phi = 238°$

B. $\phi = 58°$

C. $\phi = 122°$

Correct answer: C. $\phi = 122°$

Formulario / principles used

- Complex transfer function:

$$\overline{\Phi}(\omega) = \Re\{\overline{\Phi}\} + i\Im\{\overline{\Phi}\}$$

- Phase angle:

$$\phi = \arg\left(\overline{\Phi}\right) = \mathrm{atan2}\left(\Im\{\overline{\Phi}\}, \Re\{\overline{\Phi}\}\right)$$

- If $\Re < 0$ and $\Im > 0$, the point lies in quadrant II:

$$\phi = 180° - \tan^{-1}\left(\frac{|\Im|}{|\Re|}\right)$$

Step-by-step solution

1. Identify the signs of the complex components:

$$\Re = -5.0 \times 10^{-2} = -0.05 < 0, \qquad \Im = 8.0 \times 10^{-2} = 0.08 > 0.$$

2. Since $\Re < 0$ and $\Im > 0$, the complex number is in quadrant II.
3. Compute the reference angle:

$$\alpha = \tan^{-1}\left(\frac{|\Im|}{|\Re|}\right) = \tan^{-1}\left(\frac{0.08}{0.05}\right) = \tan^{-1}(1.6) = 58°.$$

4. Convert to the quadrant-II angle:

$$\phi = 180° - \alpha = 180° - 58° = 122°.$$

5. Therefore, the phase lag is:

$$\boxed{\phi = 122°}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

M2. Pitch RAO phase, quadrant II variation

SHIP MOTION DYNAMICS. Transfer functions of a ship are available. At a given frequency, the real and imaginary parts of the pitch RAO are $(\mathrm{Re}, \mathrm{Im}) = (-4.0 \times 10^{-2},\ 3.0 \times 10^{-2})$ rad/m. What is the phase angle ε of the complex RAO, expressed in degrees in the interval $[0, 360)$?

Options:

A. 36.9°

B. 143.1°

C. 216.9°

D. 323.1°

Correct answer: B. 143.1°

Formulario / principles used

$$\varepsilon = \operatorname{atan2}\left(\Im\{\overline{\Phi}\}, \Re\{\overline{\Phi}\}\right)$$

For quadrant II:

$$\varepsilon = 180^\circ - \tan^{-1}\left(\frac{|\Im|}{|\Re|}\right)$$

Step-by-step solution

1. Insert the given complex components:

$$\Re = -4.0 \times 10^{-2} = -0.04, \qquad \Im = 3.0 \times 10^{-2} = 0.03.$$

2. The point is in quadrant II because:

$$\Re < 0, \qquad \Im > 0.$$

3. Compute the reference angle:

$$\alpha = \tan^{-1}\left(\frac{0.03}{0.04}\right) = \tan^{-1}(0.75) = 36.9^\circ.$$

4. Convert to quadrant II:

$$\varepsilon = 180^\circ - 36.9^\circ = 143.1^\circ.$$

5. Therefore:

$$\boxed{\varepsilon = 143.1^\circ}.$$

Source(s): 41107_mock_exam_with_answers / Block 4 reference PDF

M3. Heave natural frequency of a floating cylinder

SHIP MOTION DYNAMICS: A vertical circular cylinder floats with its longitudinal axis vertically aligned. The cylinder undergoes free oscillations (pure heave); damping is neglected. The cylinder has diameter $D = 0.3$ m and floats at a draught $h = 1.1$ m measured from the bottom of the cylinder, when at rest in still water. At which frequency ω_N does the cylinder oscillate?

Options:

A. $\omega_N > 3$ rad/s

B. $\omega_N = 3$ rad/s

C. $\omega_N < 3$ rad/s

Correct answer: C. $\omega_N < 3$ rad/s

Formulario / principles used

- Heave equation without damping:

$$(m + a_{33})\ddot{z} + C_{33}z = 0$$

- Natural angular frequency:

$$\omega_N = \sqrt{\frac{C_{33}}{m + a_{33}}}$$

- For a vertical cylinder, ignoring added mass:

$$C_{33} = \rho g A_w, \qquad m = \rho A_w h$$

$$\omega_{N,0} = \sqrt{\frac{\rho g A_w}{\rho A_w h}} = \sqrt{\frac{g}{h}}$$

- Added mass increases the denominator, so:

$$\omega_N < \omega_{N,0}$$

Step-by-step solution

1. Compute the no-added-mass estimate:

$$\omega_{N,0} = \sqrt{\frac{g}{h}} = \sqrt{\frac{9.81}{1.1}}.$$

2. Calculate the value inside the square root:

$$\frac{9.81}{1.1} = 8.918.$$

3. Take the square root:

$$\omega_{N,0} = \sqrt{8.918} = 2.99 \text{ rad/s}.$$

4. Even without added mass, the result is slightly below 3 rad/s:

$$2.99 \text{ rad/s} < 3 \text{ rad/s}.$$

5. With added mass, the denominator becomes $m + a_{33} > m$, so the frequency becomes even smaller:

$$\omega_N = \sqrt{\frac{C_{33}}{m + a_{33}}} < \sqrt{\frac{C_{33}}{m}}.$$

6. Therefore:

$$\boxed{\omega_N < 3 \text{ rad/s}}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

M4. Heave natural frequency, changed draught

SHIP MOTION DYNAMICS. A vertical circular cylinder floats with its axis vertical and undergoes free oscillations in pure heave. Damping is neglected. The draught measured from the bottom of the cylinder is $h = 0.80$ m. What is the natural angular frequency approximately?

Options:

A. 2.5 rad/s

B. 3.5 rad/s

C. 4.5 rad/s

D. 5.5 rad/s

Correct answer: B. 3.5 rad/s

Formulario / principles used

$$\omega_n = \sqrt{\frac{g}{h}}$$

where h is the draught of the vertical floating cylinder.

Step-by-step solution

1. Insert the given draught:
$$\omega_n = \sqrt{\frac{9.81}{0.80}}.$$
2. Compute the ratio:
$$\frac{9.81}{0.80} = 12.2625.$$
3. Take the square root:
$$\omega_n = \sqrt{12.2625} = 3.50 \text{ rad/s}.$$
4. Therefore:
$$\boxed{\omega_n \approx 3.5 \text{ rad/s}}.$$

Source(s): 41107_mock_exam_with_answers / Block 4 reference PDF

M5. Encountered period to true wave period

WAVES: A ship sails with a speed of 18 knots. The ship encounters regular waves at an angle $\beta = 135°$ (bow-quartering waves). On the ship, the encountered wave period is $T_{\text{enc}} = 10$ s. What is the absolute (true) wave period T, as measured by a fixed observer relative to the waves?

Options:

A. $T = 12.5$ s

B. $T = 13.0$ s

C. $T = 13.5$ s

Correct answer: B. $T = 13.0$ s

Formulario / principles used

- Speed conversion:
$$U[\mathrm{m/s}] = 0.51444\,U[\mathrm{knots}]$$
- Encounter angular frequency:
$$\omega_e = \frac{2\pi}{T_{\mathrm{enc}}}$$
- Deep-water encounter equation:
$$\omega_e = \omega - \frac{\omega^2 U}{g}\cos\beta$$
- True wave period:
$$T = \frac{2\pi}{\omega}$$

Step-by-step solution

1. Convert ship speed:
$$U = 18(0.51444) = 9.26 \text{ m/s}.$$
2. Calculate encountered angular frequency:
$$\omega_e = \frac{2\pi}{10} = 0.628 \text{ rad/s}.$$
3. Evaluate heading cosine:
$$\cos(135^\circ) = -0.707.$$
4. Substitute into the encounter equation:
$$0.628 = \omega - \frac{\omega^2(9.26)}{9.81}(-0.707).$$
5. Since $\cos(135^\circ) < 0$, the quadratic term is positive:
$$0.628 = \omega + 0.667\omega^2.$$
6. Rearrange:
$$0.667\omega^2 + \omega - 0.628 = 0.$$
7. Solve the quadratic equation and keep the positive root:
$$\omega \approx 0.477 \text{ rad/s}.$$
8. Convert to true wave period:
$$T = \frac{2\pi}{0.477} = 13.18 \text{ s}.$$
9. Closest option:
$$\boxed{T \approx 13.0 \text{ s}}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

M6. Encountered period in head seas, 20 knots

WAVE THEORY - Encountered Period (Head Seas). A ship travels at 20 knots in head seas ($\beta = 180^\circ$). The waves have frequency $f = 0.1$ Hz. What is the encountered period T_e?

Options:

A. $T_e = 8.2$ s

B. $T_e = 6.9$ s

C. $T_e = 6.0$ s

Correct answer: C. $T_e = 6.0$ s

Formulario / principles used

$$\omega = 2\pi f$$

$$\omega_e = \omega - \frac{\omega^2 U}{g}\cos\beta$$

For head seas, $\beta = 180°$, so $\cos\beta = -1$:

$$\omega_e = \omega + \frac{\omega^2 U}{g}$$

$$T_e = \frac{2\pi}{\omega_e}$$

Step-by-step solution

1. Convert ship speed:
$$U = 20(0.51444) = 10.29 \text{ m/s}.$$

2. Convert true wave frequency to angular frequency:
$$\omega = 2\pi(0.1) = 0.628 \text{ rad/s}.$$

3. Compute the forward-speed correction:
$$\frac{\omega^2 U}{g} = \frac{(0.628)^2(10.29)}{9.81} = 0.414 \text{ rad/s}.$$

4. Add the correction for head seas:
$$\omega_e = 0.628 + 0.414 = 1.042 \text{ rad/s}.$$

5. Convert to encountered period:
$$T_e = \frac{2\pi}{1.042} = 6.03 \text{ s}.$$

6. Therefore:
$$\boxed{T_e \approx 6.0 \text{ s}}.$$

Source(s): 41107_MCQ_Study_Guide / Block 4 reference PDF

M7. Encountered period in head seas, 18 knots

FORWARD SPEED / ENCOUNTERED WAVES. A ship travels at 18 knots in head seas. The true wave frequency is $f = 0.10$ Hz. What is the encountered period T_e approximately?

Options:

A. 6.3 s

B. 7.2 s

C. 8.5 s

D. 10.0 s

Correct answer: **A.** 6.3 s

Formulario / principles used

$$U = 0.51444\, U_{\text{knots}}, \qquad \omega = 2\pi f$$

$$\omega_e = \omega + \frac{\omega^2 U}{g} \quad \text{for head seas}$$

$$T_e = \frac{2\pi}{\omega_e}$$

Step-by-step solution

1. Convert speed:
$$U = 18(0.51444) = 9.26 \text{ m/s}.$$
2. Convert frequency:
$$\omega = 2\pi(0.10) = 0.628 \text{ rad/s}.$$
3. Compute the head-seas encounter frequency:
$$\omega_e = 0.628 + \frac{(0.628)^2(9.26)}{9.81}.$$
4. Evaluate the correction term:
$$\frac{(0.628)^2(9.26)}{9.81} = 0.373 \text{ rad/s}.$$
5. Thus:
$$\omega_e = 0.628 + 0.373 = 1.001 \text{ rad/s}.$$
6. Convert to encountered period:
$$T_e = \frac{2\pi}{1.001} = 6.28 \text{ s}.$$
7. Closest option:
$$\boxed{T_e \approx 6.3 \text{ s}}.$$

Source(s): 41107_mock_exam_with_answers / Block 4 reference PDF

M8. Froude-Krylov assumption in ship motion theory

FROUDE-KRYLOV ASSUMPTION - Ship Motion Theory. In ship motion dynamics, the Froude-Krylov force is calculated under which assumption?

Options:

A. Wave diffraction always dominates the exciting force.

B. The presence of the hull has no effect on waves.

C. Wave diffraction and radiation are neglected.

Correct answer: B. The presence of the hull has no effect on waves.

Formulario / principles used

- Total excitation is often understood as:

$$F_{\text{exc}} = F_{\text{FK}} + F_{\text{diff}}$$

- Froude-Krylov force is calculated from the undisturbed incident-wave pressure:

$$F_{\text{FK}} = \int_S p_{\text{incident}}\, \mathbf{n}\, dS$$

- Diffraction accounts for how the body modifies the incident wave field.

Step-by-step solution

1. The Froude-Krylov force uses the pressure field from the incoming wave.
2. This pressure field is the one that would exist if the body were not present.
3. Therefore the body/hull is assumed not to disturb the wave field.
4. The effect of the body disturbing the wave field is diffraction, which is not part of the pure Froude-Krylov contribution.
5. Hence the correct statement is:

$$\boxed{\text{the presence of the hull has no effect on waves.}}$$

Source(s): Test_Exam_all extra MCQ / 41107_MCQ_Study_Guide / Block 4 reference PDF

Ship hydrostatics, stability, trim and hydrostatic coefficients

H1. Draught for transverse metacentre in the waterplane

SHIP HYDROSTATICS: A rectangular, box-shaped barge of length $L = 50$ m and beam $B = 7$ m floats at a uniform draught T. What must be the draught for the transverse metacentre (M_T) to lie in the waterplane?

Options:

A. $T = 2.7$ m

B. $T = 2.9$ m

C. $T = 3.1$ m

Correct answer: B. $T = 2.9$ m

Formulario / principles used For a rectangular box barge:

$$KB = \frac{T}{2}$$

$$I_T = \frac{LB^3}{12}, \qquad \nabla = LBT$$

$$BM_T = \frac{I_T}{\nabla} = \frac{LB^3/12}{LBT} = \frac{B^2}{12T}$$

$$KM_T = KB + BM_T = \frac{T}{2} + \frac{B^2}{12T}$$

If M_T lies in the waterplane:

$$KM_T = T$$

Step-by-step solution

1. Apply the condition $KM_T = T$:
$$\frac{T}{2} + \frac{B^2}{12T} = T.$$
2. Move $T/2$ to the right-hand side:
$$\frac{B^2}{12T} = \frac{T}{2}.$$
3. Multiply by $12T$:
$$B^2 = 6T^2.$$
4. Solve for T:
$$T = \frac{B}{\sqrt{6}}.$$
5. Substitute $B = 7$ m:
$$T = \frac{7}{\sqrt{6}} = \frac{7}{2.449} = 2.86 \text{ m}.$$
6. Closest option:
$$\boxed{T = 2.9 \text{ m}}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

H2. Draught for transverse metacentre, changed beam

SHIP HYDROSTATICS. A rectangular, box-shaped barge has beam $B = 8.0$ m and floats at uniform draught T. What must the draught be for the transverse metacentre to lie in the waterplane?

Options:

A. $T = 2.5$ m

B. $T = 3.3$ m

C. $T = 4.0$ m

D. $T = 4.6$ m

Correct answer: B. $T = 3.3$ m

Formulario / principles used Using the same rectangular-barge condition:

$$T = \frac{B}{\sqrt{6}}$$

Step-by-step solution

1. Insert the beam:
$$T = \frac{8.0}{\sqrt{6}}.$$
2. Evaluate:
$$\sqrt{6} = 2.449, \qquad T = \frac{8.0}{2.449} = 3.27 \text{ m}.$$
3. Closest option:
$$\boxed{T = 3.3 \text{ m}}.$$

Source(s): 41107_mock_exam_with_answers / Block 5 reference PDF

H3. Metacentric height from righting lever

SHIP HYDROSTATICS - Metacentric Height GM_T from GZ. A ship heels $\phi = 5°$ and has a righting lever $GZ = 0.5$ m. What is the transverse metacentric height GM_T?

Options:

A. $GM_T = 4.5$ m

B. $GM_T = 5.3$ m

C. $GM_T = 5.7$ m

Correct answer: C. $GM_T = 5.7$ m

Formulario / principles used For small-angle stability:

$$GZ = GM_T \sin\phi$$

Therefore:

$$GM_T = \frac{GZ}{\sin\phi}$$

Step-by-step solution

1. Rearrange the righting lever relation:
$$GM_T = \frac{GZ}{\sin\phi}.$$
2. Substitute the numerical values:
$$GM_T = \frac{0.5}{\sin(5^\circ)}.$$
3. Calculate:
$$\sin(5^\circ) = 0.08716.$$
4. Therefore:
$$GM_T = \frac{0.5}{0.08716} = 5.74 \text{ m}.$$
5. Closest option:
$$\boxed{GM_T = 5.7 \text{ m}}.$$

Source(s): Test_Exam_all extra MCQ / 41107_MCQ_Study_Guide / Block 5 reference PDF

H4. Stability of a floating square log

SHIP HYDROSTATICS: A homogeneous log of square cross section, side $b = 1$ m, and length $l = 6$ m long floats with its length parallel to the water surface and with its cross-sectional sides horizontally and vertically aligned. The log's density is half that of the water. What happens to the log if it is given a small heeling moment which is instantly applied and released? Stillwater conditions are assumed.

Options:

A. The log is stable in its initial equilibrium position and will return to this position after the heeling moment is applied and released.

B. The log is in an unstable equilibrium in its initial equilibrium position and will find a new equilibrium position after the heeling moment is applied and released.

C. The log remains in the heeled position that it reaches after the heeling moment is applied and released since the log is unconditionally stable.

Correct answer: B. The log is initially unstable and will find a new equilibrium position.

Formulario / principles used

$$GM_T = KB + BM_T - KG$$

For the square log:
$$T = \frac{b}{2}, \qquad KB = \frac{T}{2}, \qquad BM_T = \frac{b^2}{12T}, \qquad KG = \frac{b}{2}$$

Stability rule:
$$GM_T > 0 \Rightarrow \text{stable}, \qquad GM_T < 0 \Rightarrow \text{unstable}$$

Step-by-step solution

1. Since the log density is half the water density, half of the square section is submerged:

$$T = \frac{b}{2} = \frac{1}{2} = 0.5 \text{ m}.$$

2. Compute the centre of buoyancy:

$$KB = \frac{T}{2} = \frac{0.5}{2} = 0.25 \text{ m} = \frac{b}{4}.$$

3. Compute the metacentric radius:

$$BM_T = \frac{b^2}{12T} = \frac{1^2}{12(0.5)} = \frac{1}{6} = 0.167 \text{ m} = \frac{b}{6}.$$

4. For a homogeneous square log, the centre of gravity is at the geometric centre:

$$KG = \frac{b}{2} = 0.5 \text{ m}.$$

5. Compute transverse metacentric height:

$$GM_T = KB + BM_T - KG = 0.25 + 0.167 - 0.5 = -0.083 \text{ m}.$$

6. Since:

$$GM_T < 0,$$

the initial upright equilibrium is unstable.

7. Therefore, after being disturbed, the log will move to another equilibrium orientation.

Source(s): Test MC exam_Corr / Block 5 reference PDF

H5. Displacement of a rectangular box barge

SHIP HYDROSTATICS - Equilibrium and Buoyancy. A rectangular box barge ($L = 50$ m, $B = 8$ m) floats in freshwater ($\rho = 1000$ kg/m^3) at draught $T = 4$ m. What is the displacement in metric tons?

Options:

A. $1,000$ tons

B. $1,600$ tons

C. $2,000$ tons

Correct answer: B. $1,600$ tons

Formulario / principles used

$$\nabla = LBT$$

$$\Delta = \rho\nabla$$

$$1 \text{ metric ton} = 1000 \text{ kg}$$

where ∇ is displaced volume and Δ is displacement mass.

Step-by-step solution

1. Compute displaced volume:

$$\nabla = LBT = (50)(8)(4) = 1600\ \text{m}^3.$$

2. Compute displacement mass:

$$\Delta = \rho\nabla = (1000)(1600) = 1,600,000\ \text{kg}.$$

3. Convert to metric tons:

$$\Delta = \frac{1,600,000}{1000} = 1600\ \text{tons}.$$

4. Therefore:

$$\boxed{\Delta = 1600\ \text{tons}}.$$

Source(s): 41107_MCQ_Study_Guide / Block 5 reference PDF

H6. Block coefficient

HYDROSTATIC COEFFICIENTS. A ship in saltwater has $L = 120$ m, $B = 20$ m, $T = 8$ m and displacement mass $\Delta = 15000$ t. Taking $\rho = 1.025$ t/m^3, what is the block coefficient C_B approximately?

Options:

A. 0.56

B. 0.68

C. 0.76

D. 0.91

Correct answer: C. $C_B = 0.76$

Formulario / principles used

$$\nabla = \frac{\Delta}{\rho}$$

$$C_B = \frac{\nabla}{LBT}$$

where ∇ is displaced volume and LBT is the volume of the circumscribing rectangular block.

Step-by-step solution

1. Convert displacement mass to displaced volume:

$$\nabla = \frac{15000}{1.025} = 14634 \text{ m}^3.$$

2. Compute the rectangular block volume:

$$LBT = (120)(20)(8) = 19200 \text{ m}^3.$$

3. Compute the block coefficient:

$$C_B = \frac{14634}{19200} = 0.762.$$

4. Closest option:

$$\boxed{C_B \approx 0.76}.$$

Source(s): 41107_mock_exam_with_answers / Block 5 reference PDF

H7. Trim due to an added load

SHIP HYDROSTATICS - Trim and Longitudinal Stability. A 100 m barge floats on an even keel. A load of 50 tons is added at 30 m forward of midship. The barge has $GM_L = 60$ m and displacement $\Delta = 2000$ tons. What is the change in trim (bow-down positive)?

Options:

A. Δ trim $= +0.75$ m bow-down

B. Δ trim $= -0.75$ m stern-down

C. Δ trim $= +1.5$ m

Correct answer: A. $+0.75$ m **bow-down**

Note on this MCQ. The sign is unambiguous: adding weight forward makes the bow go down. A simple total-trim estimate gives about 1.25 m total trim, while the source answer reports the bow-down change as about +0.75 m. The option to select is therefore the positive bow-down one.

Formulario / principles used

- Trimming moment:

$$M_{\text{trim}} = wd$$

- Approximate total trim:

$$t = \frac{wdL}{\Delta GM_L}$$

- If the load is added forward of midship:

bow-down trim is positive.

Step-by-step solution

1. Compute the trimming moment:

$$M_{\text{trim}} = wd = (50)(30) = 1500 \text{ ton m}.$$

2. Use the simplified trim estimate:

$$t = \frac{wdL}{\Delta GM_L}.$$

3. Substitute the numerical values:

$$t = \frac{(50)(30)(100)}{(2000)(60)}.$$

4. Compute numerator and denominator:

$$(50)(30)(100) = 150000, \qquad (2000)(60) = 120000.$$

5. Thus:

$$t = \frac{150000}{120000} = 1.25 \text{ m} \quad \text{total trim estimate}.$$

6. Since the load is placed forward of midship, the bow moves down:

$$\text{sign} = + \quad \text{(bow-down)}.$$

7. Among the available answers, the correct intended option is:

$$\boxed{+0.75 \text{ m bow-down}}.$$

Source(s): Block 5 reference PDF / generated study bank

Strength of ships

S1. Neutral axis of a thin-walled barge section

STRENGTH OF SHIPS: The cross section of a barge is shown below. All plate thicknesses are $t = 10$ mm. What is the vertical distance z_n from the keel to the neutral axis?

Options:

A. $z_n = 2.72$ m

B. $z_n = 2.81$ m

C. $z_n = 2.95$ m

Correct answer: B. $z_n = 2.81$ m

Formulario / principles used

- Neutral axis for a thin-walled section:
$$z_n = \frac{\sum_i A_i z_i}{\sum_i A_i}$$
- Plate area:
$$A_i = l_i t_i$$
- If all plate thicknesses are equal ($t_i = t$), then t cancels:
$$z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}$$

Step-by-step solution

1. Split the cross-section into thin plate elements.
2. Since all plate thicknesses are equal:
$$A_i = l_i t \quad \Rightarrow \quad z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}.$$
3. From the sketch, the summed plate area is:
$$\sum_i A_i = 0.52 \text{ m}^2.$$
4. The first moment of area about the keel is:
$$\sum_i A_i z_i = 1.46 \text{ m}^3.$$
5. Compute the neutral-axis height:
$$z_n = \frac{1.46}{0.52} = 2.81 \text{ m}.$$
6. Therefore:
$$\boxed{z_n = 2.81 \text{ m}}.$$

Source(s): Test MC exam_Corr / Test_Exam_all / Block 6 reference PDF

S2. Neutral axis, explicit plate lengths

STRENGTH OF SHIPS. The thin-walled barge cross section shown below has uniform plate thickness $t = 10$ mm. What is the vertical distance z_n from the keel to the neutral axis?

Options:

A. $z_n = 2.32$ m

B. $z_n = 2.43$ m

C. $z_n = 2.57$ m

D. $z_n = 3.00$ m

Correct answer: B. $z_n = 2.43$ m

Formulario / principles used

$$z_n = \frac{\sum_i A_i z_i}{\sum_i A_i}$$

For uniform thickness:

$$z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}$$

Step-by-step solution

1. The plate lengths are:
$$l_i = (10, 10, 10, 6, 6, 1, 1) \text{ m}.$$
2. The corresponding centroid heights are:
$$z_i = (0, 1, 6, 3, 3, 0.5, 0.5) \text{ m}.$$
3. Compute the numerator:
$$\sum_i l_i z_i = (10)(0) + (10)(1) + (10)(6) + (6)(3) + (6)(3) + (1)(0.5) + (1)(0.5).$$
4. Evaluate:
$$\sum_i l_i z_i = 0 + 10 + 60 + 18 + 18 + 0.5 + 0.5 = 107.$$
5. Compute the denominator:
$$\sum_i l_i = 10 + 10 + 10 + 6 + 6 + 1 + 1 = 44.$$
6. Therefore:
$$z_n = \frac{107}{44} = 2.43 \text{ m}.$$
7. Correct option:
$$\boxed{z_n = 2.43 \text{ m}}.$$

Source(s): 41107_mock_exam_with_answers / Block 6 reference PDF

S3. Distributed load equilibrium, triangular gravel load

STRENGTH OF SHIPS: The diagram below shows the distributed weight on a rectangular, box-shaped barge loaded with gravel. The barge is made from steel and its own weight is uniformly distributed along the length with intensity $0.1p$. The barge's length and breadth are 50 m and 8 m, respectively, and it floats at the draught 4 m in freshwater (no heel and no trim). Approximately, what value has p?

Options:

A. $p = 53$ tons

B. $p = 38$ tons

C. $p = 45$ tons/m

Correct answer: A. $p = 53$ tons **(intended as** tons/m **in the load diagram)**

Formulario / principles used

- Freshwater displacement in tonnes for a box barge:

$$\Delta = LBT$$

- Triangular load resultant:

$$W_{\text{gravel}} = \frac{1}{2}Lp$$

- Uniform self-weight resultant:

$$W_{\text{steel}} = L(0.1p)$$

- Static equilibrium:

$$\Delta = W_{\text{gravel}} + W_{\text{steel}}$$

Step-by-step solution

1. Compute displaced volume:

$$\nabla = LBT = (50)(8)(4) = 1600 \text{ m}^3.$$

2. In freshwater, this corresponds approximately to:

$$\Delta = 1600 \text{ tons}.$$

3. Compute triangular gravel load:

$$W_{\text{gravel}} = \frac{1}{2}Lp = \frac{1}{2}(50)p = 25p.$$

4. Compute steel self-weight:

$$W_{\text{steel}} = L(0.1p) = (50)(0.1p) = 5p.$$

5. Total weight:

$$W_{\text{total}} = 25p + 5p = 30p.$$

6. Apply equilibrium:

$$30p = 1600.$$

7. Solve:

$$p = \frac{1600}{30} = 53.3 \text{ tons/m}.$$

8. Closest/intended option:

$$\boxed{p \approx 53 \text{ tons/m}}.$$

Source(s): Test_Exam_all / 41107_MCQ_Study_Guide / Block 6 reference PDF

S4. Distributed load equilibrium, changed dimensions

STRENGTH OF SHIPS. The barge below is loaded with gravel. The gravel load is triangular along the whole length with maximum intensity p at midship. The barge's own weight is uniformly distributed with intensity $0.2p$. The barge length is 40 m, breadth is 6 m, and it floats at draught $T = 3$ m in freshwater. Approximately what value has p?

Options:

A. $p = 18$ tons/m

B. $p = 26$ tons/m

C. $p = 32$ tons/m

D. $p = 40$ tons/m

Correct answer: B. $p = 26$ tons/m

Formulario / principles used

$$\Delta = LBT$$

$$W_{\text{gravel}} = \frac{1}{2}Lp, \qquad W_{\text{own}} = L(0.2p)$$

$$LBT = \frac{1}{2}Lp + L(0.2p)$$

Step-by-step solution

1. Compute freshwater displacement mass:

$$\Delta = LBT = (40)(6)(3) = 720 \text{ tons.}$$

2. Compute triangular gravel load:

$$W_{\text{gravel}} = \frac{1}{2}(40)p = 20p.$$

3. Compute barge own weight:

$$W_{\text{own}} = (40)(0.2p) = 8p.$$

4. Total load:

$$W_{\text{total}} = 20p + 8p = 28p.$$

5. Equilibrium:

$$28p = 720.$$

6. Solve for p:

$$p = \frac{720}{28} = 25.7 \text{ tons/m.}$$

7. Closest option:

$$\boxed{p \approx 26 \text{ tons/m}}.$$

Source(s): 41107_mock_exam_with_answers / Block 6 reference PDF

S5. Location of greatest bending stress

STRENGTH OF SHIPS - Greatest Bending Stress Location. A barge cross-section has identical plate thicknesses throughout and has an inner bottom plate creating an asymmetric section. The neutral axis is NOT at midheight. Which location experiences the greatest bending stress?

Options:

A. The plate at deck level

B. The plate at keel level

C. Cannot be determined without material properties

Correct answer: A. The plate at deck level

Formulario / principles used

$$\sigma = \frac{My}{I}$$

where M is bending moment, I is second moment of area, and y is distance from the neutral axis. Largest absolute bending stress:

$$|\sigma|_{\max} \quad \text{occurs where} \quad |y| \text{ is maximum.}$$

Step-by-step solution

1. Bending stress varies linearly with distance from the neutral axis:

$$\sigma \propto y.$$

2. Material properties are not needed to locate the maximum stress in this homogeneous section.
3. The section is asymmetric, and the neutral axis is closer to the keel than to the deck.
4. Therefore:

$$y_{\text{deck}} > y_{\text{keel}}.$$

5. Since:

$$|\sigma| = \left|\frac{My}{I}\right|,$$

the largest stress occurs at the largest $|y|$.

6. Hence the deck plate experiences the greatest bending stress.

Source(s): 41107_MCQ_Study_Guide / Block 6 reference PDF

S6. Deck bending stress from section modulus relation

SHIP STRENGTH - Second Moment of Area (Section Modulus). A simplified ship cross-section has a deck plate at $z = 10$ m and keel at $z = 0$, neutral axis at $z_n = 4$ m. The second moment of area is $I = 20$ m^4. For a sagging moment $M = 500$ MNm, what is the deck stress?

Options:

A. $\sigma_{\text{deck}} = 150$ MPa

B. $\sigma_{\text{deck}} = 100$ MPa

C. $\sigma_{\text{deck}} = 60$ MPa

Correct answer: A. $\sigma_{\text{deck}} = 150$ MPa

Formulario / principles used

$$\sigma = \frac{My}{I}$$

$$y_{\text{deck}} = z_{\text{deck}} - z_n$$

$$1 \text{ MPa} = 10^6 \text{ Pa}, \qquad 1 \text{ MNm} = 10^6 \text{ Nm}$$

Step-by-step solution

1. Find the distance from the neutral axis to the deck:
$$y_{\text{deck}} = z_{\text{deck}} - z_n = 10 - 4 = 6 \text{ m}.$$
2. Convert bending moment:
$$M = 500 \text{ MNm} = 500 \times 10^6 \text{ Nm}.$$
3. Apply bending stress formula:
$$\sigma_{\text{deck}} = \frac{M y_{\text{deck}}}{I}.$$
4. Substitute numerical values:
$$\sigma_{\text{deck}} = \frac{(500 \times 10^6)(6)}{20}.$$
5. Evaluate:
$$\sigma_{\text{deck}} = 150 \times 10^6 \text{ Pa}.$$
6. Convert to MPa:
$$150 \times 10^6 \text{ Pa} = 150 \text{ MPa}.$$
7. Therefore:
$$\boxed{\sigma_{\text{deck}} = 150 \text{ MPa}}.$$

Source(s): 41107_MCQ_Study_Guide / Block 6 reference PDF

Ship resistance, model testing and Froude scaling

R1. Similarity condition in towing-tank resistance estimation

SHIP RESISTANCE / MODEL TESTING. In towing-tank resistance estimation for ships with free-surface effects, what similarity must be preserved in practice?

Options:

A. Identical Froude numbers of model and ship

B. Identical Reynolds numbers of model and ship

C. Identical Froude and Reynolds numbers simultaneously

D. Identical Strouhal numbers only

Correct answer: A. Identical Froude numbers of model and ship

Formulario / principles used

$$Fn = \frac{U}{\sqrt{gL}}$$

$$Re = \frac{UL}{\nu}$$

Free-surface wave-making similarity is governed primarily by Fn. Exact Fn and Re similarity cannot generally be achieved simultaneously in the same fluid.

Step-by-step solution

1. Ship resistance includes wave-making and viscous/frictional components.
2. Wave-making depends strongly on the Froude number:
$$Fn = \frac{U}{\sqrt{gL}}.$$
3. Therefore, model and ship are compared at corresponding speeds such that:
$$Fn_m = Fn_s.$$
4. Reynolds number similarity would require:
$$Re_m = Re_s,$$
but this cannot be achieved at the same time as Froude similarity in normal same-fluid model tests.
5. Therefore the preserved similarity is:
$$\boxed{Fn_m = Fn_s}.$$

Source(s): 41107_mock_exam_with_answers / Block 7 reference PDF

R2. Resistance estimation via model tests

SHIP RESISTANCE: In resistance estimation via model tests, it is important to ensure what?

Options:

A. Identical Reynolds numbers as well as Froude numbers of the full-scale and model-scale ships.

B. Identical Froude numbers of the full-scale and model-scale ships.

C. Identical Reynolds numbers of the full-scale and model-scale ships.

Correct answer: B. Identical Froude numbers of the full-scale and model-scale ships.

Formulario / principles used

$$Fn = \frac{U}{\sqrt{gL}}, \qquad Re = \frac{UL}{\nu}$$

$Fn_m = Fn_s$ is the practical model-test condition for free-surface resistance.

Step-by-step solution

1. For a geometrically scaled model, the model length is much smaller than the ship length.
2. If Froude number is matched, the model speed follows:
$$\frac{U_s}{U_m} = \sqrt{\frac{L_s}{L_m}}.$$
3. This does not also make Reynolds number equal, because:
$$Re = \frac{UL}{\nu}$$
depends on both speed and length.
4. Since wave-making is governed by Froude number, the practical condition is:
$$Fn_s = Fn_m.$$
5. Viscous effects are then corrected separately.

Source(s): Test MC exam_Corr / Block 7 reference PDF

R3. Resistance ratio at corresponding speeds

SHIP RESISTANCE: A ship of mass $5,000$ tons, length 120 m, is to be represented by a model 3 m long for resistance estimation. If performance is compared at the corresponding speeds, what will the ratio r of ship to model resistance approximately be?

Options:

A. $r = 6 \times 10^4$

B. $r = 6 \times 10^3$

C. $r = 6 \times 10^5$

Correct answer: A. $r = 6 \times 10^4$

Formulario / principles used

$$\alpha = \frac{L_s}{L_m}$$

At corresponding Froude speeds, force/resistance scales approximately as volume:

$$\frac{R_s}{R_m} \approx \alpha^3$$

Step-by-step solution

1. Compute the geometric scale factor:

$$\alpha = \frac{L_s}{L_m} = \frac{120}{3} = 40.$$

2. At corresponding Froude speeds:

$$\frac{R_s}{R_m} \approx \alpha^3.$$

3. Substitute:

$$\frac{R_s}{R_m} \approx 40^3.$$

4. Evaluate:

$$40^3 = 40 \cdot 40 \cdot 40 = 64000 = 6.4 \times 10^4.$$

5. Closest option:

$$\boxed{r \approx 6 \times 10^4}.$$

Source(s): Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide / Block 7 reference PDF

R4. Resistance ratio, changed model length

SHIP RESISTANCE. A ship of length 180 m is represented by a model 4.0 m long for resistance estimation. If performance is compared at corresponding speeds, what is the ratio R_s/R_m approximately?

Options:

A. 9.1×10^3

B. 9.1×10^4

C. 9.1×10^5

D. 4.5×10^4

Correct answer: B. 9.1×10^4

Formulario / principles used

$$\alpha = \frac{L_s}{L_m}$$

$$\frac{R_s}{R_m} \approx \alpha^3$$

Step-by-step solution

1. Compute length scale:
$$\alpha = \frac{180}{4.0} = 45.$$
2. Apply resistance scaling:
$$\frac{R_s}{R_m} \approx 45^3.$$
3. Evaluate:
$$45^3 = (45)(45)(45) = 91125.$$
4. Scientific notation:
$$91125 = 9.11 \times 10^4.$$
5. Closest option:
$$\boxed{\frac{R_s}{R_m} \approx 9.1 \times 10^4}.$$

Source(s): 41107_mock_exam_with_answers / Block 7 reference PDF

R5. Model weight for Froude scaling

SHIP RESISTANCE - Model Weight for Froude Scaling. A 5,000-ton ship ($L = 120$ m) is modelled at $L_m = 3.5$ m. What is the approximate weight of the model?

Options:

A. 75 kg

B. 125 kg

C. 175 kg

Correct answer: B. 125 kg

Formulario / principles used

$$\alpha = \frac{L_s}{L_m}$$

Mass and weight scale with volume:

$$\frac{m_s}{m_m} = \alpha^3$$

Therefore:

$$m_m = \frac{m_s}{\alpha^3}$$

Step-by-step solution

1. Convert ship mass:
$$m_s = 5000 \text{ tons} = 5{,}000{,}000 \text{ kg}.$$
2. Compute length scale:
$$\alpha = \frac{120}{3.5} = 34.29.$$
3. Compute volume/mass scale:
$$\alpha^3 = (34.29)^3 \approx 40{,}300.$$
4. Compute model mass:
$$m_m = \frac{5{,}000{,}000}{40{,}300} = 124 \text{ kg}.$$
5. Closest option:
$$\boxed{m_m \approx 125 \text{ kg}}.$$

Source(s): Test_Exam_all extra MCQ / 41107_MCQ_Study_Guide / Block 7 reference PDF

General formula sheet for this Ships block

Ship motion dynamics and encounter frequency

$$\overline{\Phi}(\omega) = \Re\{\overline{\Phi}\} + i\Im\{\overline{\Phi}\}$$

$$|\Phi(\omega)| = \sqrt{\left(\Re\{\overline{\Phi}\}\right)^2 + \left(\Im\{\overline{\Phi}\}\right)^2}$$

$$\epsilon = \text{atan2}\left(\Im\{\overline{\Phi}\}, \Re\{\overline{\Phi}\}\right)$$

$$\omega_e = \omega - \frac{\omega^2 U}{g}\cos\beta, \qquad T_e = \frac{2\pi}{\omega_e}$$

$$U[\text{m/s}] = 0.51444\,U[\text{knots}]$$

$$\omega_N = \sqrt{\frac{C_{33}}{m + a_{33}}}, \qquad \omega_{N,0} = \sqrt{\frac{g}{h}}$$

Hydrostatics, stability and trim

$$\nabla = LBT, \qquad \Delta = \rho\nabla$$

$$KB = \frac{T}{2}, \qquad I_T = \frac{LB^3}{12}, \qquad BM_T = \frac{I_T}{\nabla} = \frac{B^2}{12T}$$

$$KM_T = KB + BM_T, \qquad GM_T = KM_T - KG$$

$$GZ = GM_T \sin\phi, \qquad GM_T = \frac{GZ}{\sin\phi}$$

$$KM_T = T \quad \Rightarrow \quad T = \frac{B}{\sqrt{6}} \quad \text{for rectangular barge with } M_T \text{ in waterplane}$$

$$C_B = \frac{\nabla}{LBT} = \frac{\Delta/\rho}{LBT}$$

$$t \approx \frac{wdL}{\Delta GM_L}$$

Strength of ships

$$z_n = \frac{\sum_i A_i z_i}{\sum_i A_i}$$

For uniform plate thickness:

$$z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}$$

$$W_{\text{triangular}} = \frac{1}{2}Lp, \qquad W_{\text{uniform}} = Lq$$

$$\text{Static equilibrium: } W_{\text{total}} = B_{\text{buoyancy}}$$

$$\sigma = \frac{My}{I}, \qquad Z = \frac{I}{y_{\max}}, \qquad \sigma_{\max} = \frac{M}{Z}$$

Largest bending stress occurs at the largest distance from the neutral axis:

$$|y| = \max.$$

Ship resistance and model testing

$$Fn = \frac{U}{\sqrt{gL}}, \qquad Re = \frac{UL}{\nu}$$

$$R = \frac{1}{2}\rho U^2 C_T S$$

$$\alpha = \frac{L_s}{L_m}$$

$$\frac{U_s}{U_m} = \sqrt{\alpha}, \qquad \frac{S_s}{S_m} = \alpha^2, \qquad \frac{m_s}{m_m} = \alpha^3$$

$$\frac{R_s}{R_m} \approx \alpha^3$$

For ship resistance model tests with free-surface effects:

$$Fn_s = Fn_m \quad \text{is preserved in practice.}$$